

A COMPUTATIONAL MODEL FOR TUMOUR-STROMAL INTERACTIONS IN 3D CULTURE SYSTEMS

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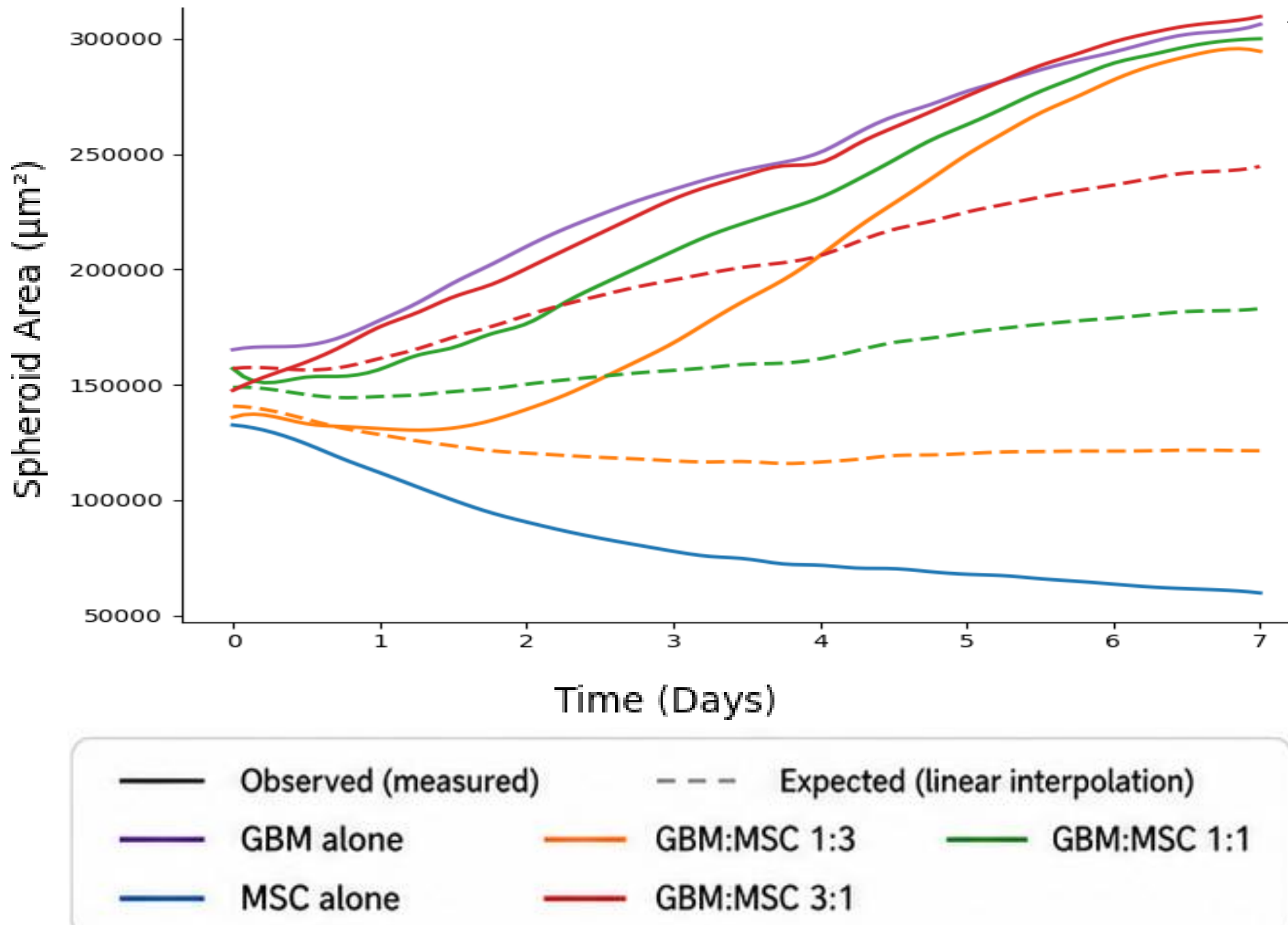


1 EXPERIMENTAL OBSERVATION: MIXED SPHEROIDS GROW NON-LINEARLY

MOTIVATION & CONTEXT

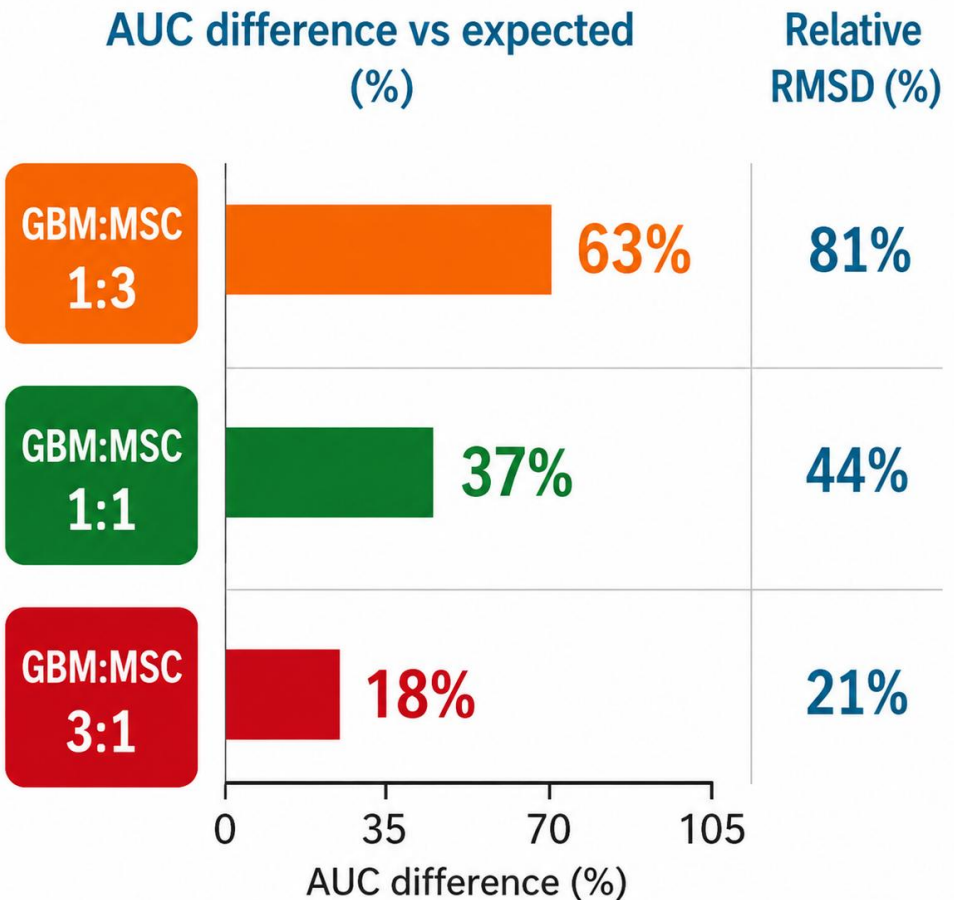
- Tumor growth is shaped not only by cancer cells, but also by interactions with the surrounding microenvironment.
- 3D GBM spheroids better mimic tumor-like organization than 2D cultures by allowing cell-cell contact, nutrient/metabolite gradients, and stressed inner regions.
- Mesenchymal stromal cells (MSCs) are context-dependent: they can either suppress or support GBM growth depending on local conditions.
- To understand MSC effects, we need to look beyond overall spheroid size and examine spatial and dynamic processes inside the spheroid.

Observed vs. linearly interpolated growth curves



Mixed **GBM-MSC** spheroids grew larger than expected from pure **GBM** and pure **MSC** controls, indicating a non-additive stromal effect.

Magnitude of non-linearity



METHODS / GROWTH ASSAY

- U87MG glioblastoma + mesenchymal stromal cell (MSC) spheroids
- 10,000 cells seeded per spheroid
- Conditions: GBM only, MSC only, GBM:MSC 3:1, 1:1, 1:3
- 7-day live imaging of spheroid growth
- Spheroid area quantified from brightfield images



What internal mechanisms could produce this non-linear, pro-tumor MSC effect?

2 OUR APPROACH: A HYBRID 3D COMPUTATIONAL MODEL

OVERVIEW

- We developed a hybrid 3D agent-based model of GBM-MSC spheroid growth that couples:
- Cellular Potts Model (CPM) for cell mechanics
 - Reaction-Diffusion fields for nutrients, metabolites, ECM components and signals
 - Probabilistic logic-based rules for cell behavior and fate decisions

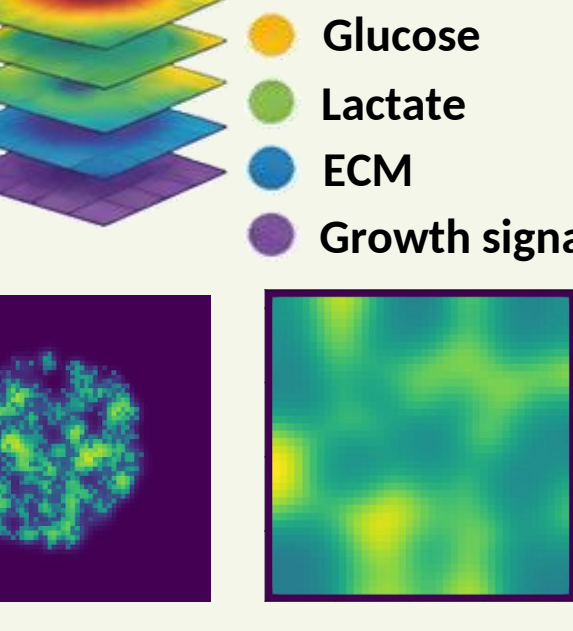
CELLS

- Cells are discrete agents
 - Evolves by minimizing an energy function (Hamiltonian)
 - Cells types & states:
- $H = \sum_{(i,j)} J(\tau(\sigma_i), \tau(\sigma_j)) \cdot (1 - \delta_{\sigma_i, \sigma_j}) + \lambda_V (V - V_{target})^2 + \lambda_S (S - S_{target})^2$
- adhesion volume surface
- GBM: P, Q, N; GSC: P, Q; MSC: anti, pro
- P = proliferative, Q = quiescent, N = necrotic

MODEL LOGIC

FIELDS

- Reaction-diffusion equations

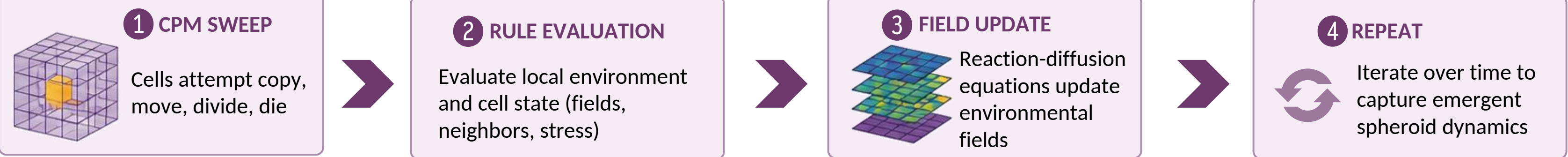


- Actions:
- diffuse
 - decay
 - consume
 - produce
 - inject

RULES

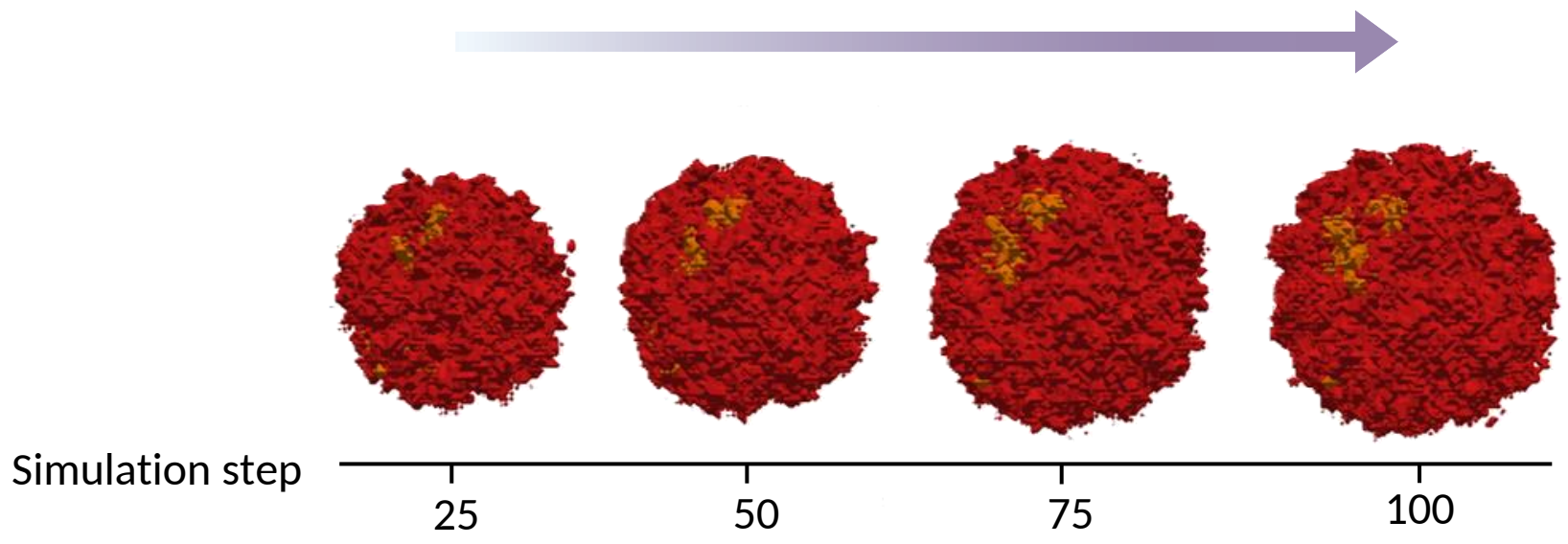
- Local sensing of environment conditions
- Probabilistic state transitions
- Cell-cycle and division control
- Context-dependent MSC phenotype switching

SIMULATION WORKFLOW

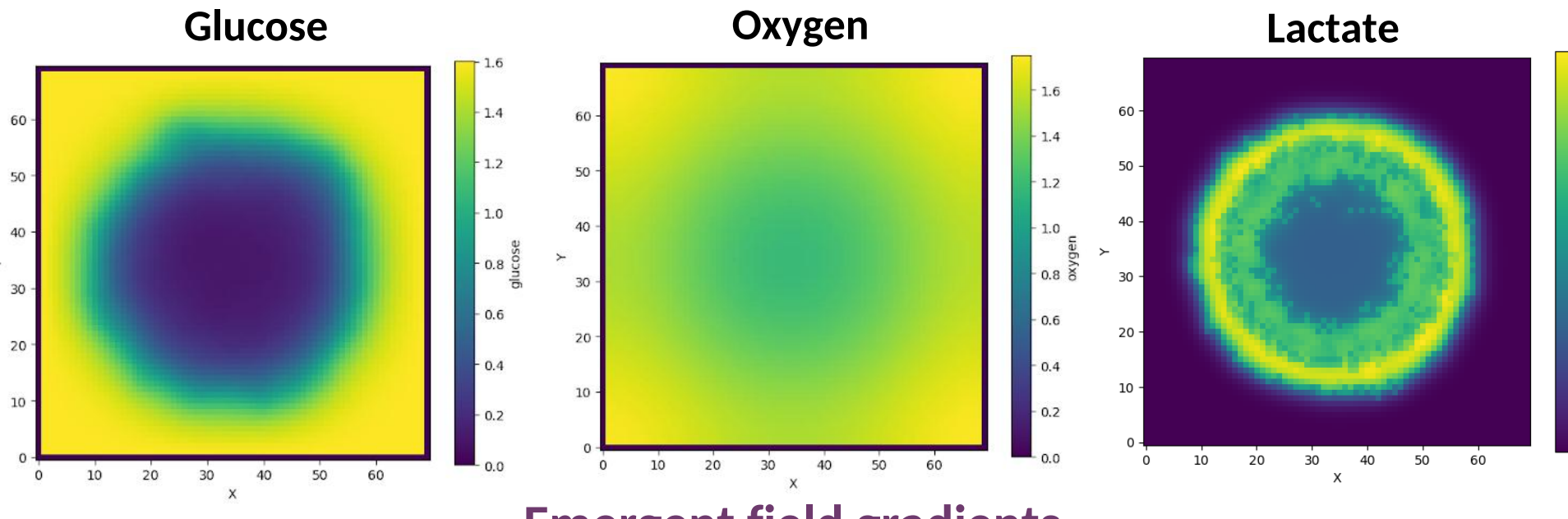


3 THE MODEL RECAPITALUTES HALLMARK FEATURES OF GBM SPHEROIDS

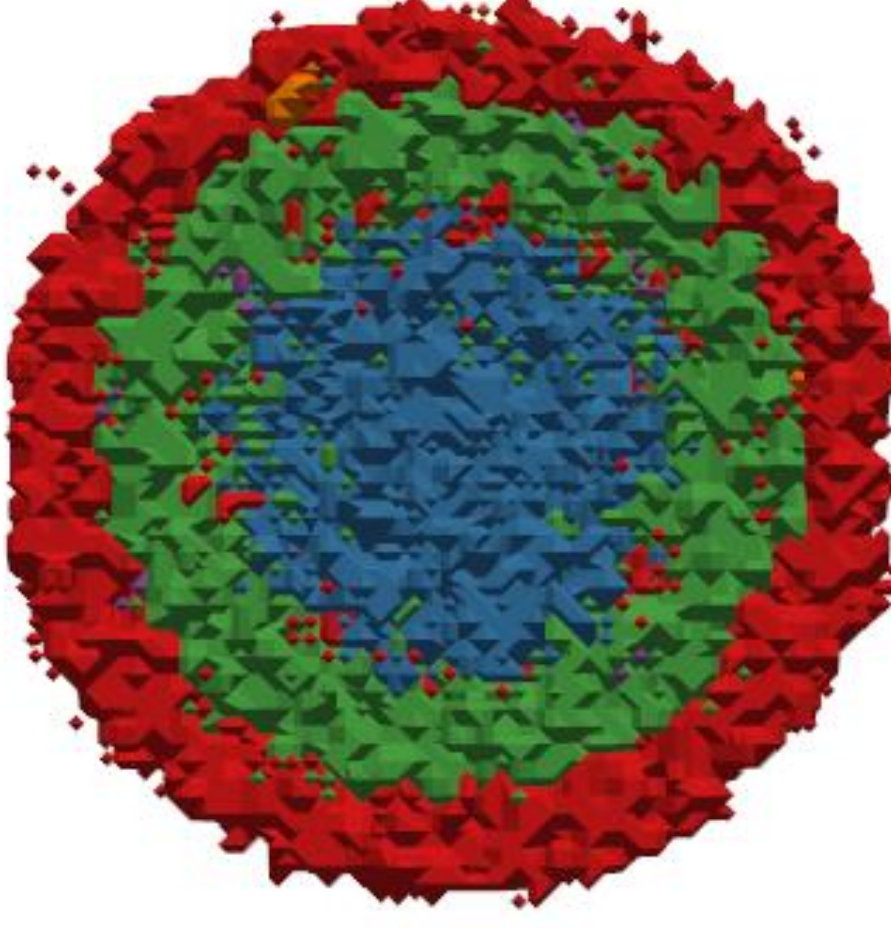
Emergent spheroid organization: nutrient gradients restrict proliferation to the outer rim and drive formation of quiescent and necrotic inner regions.



Spheroid expansion



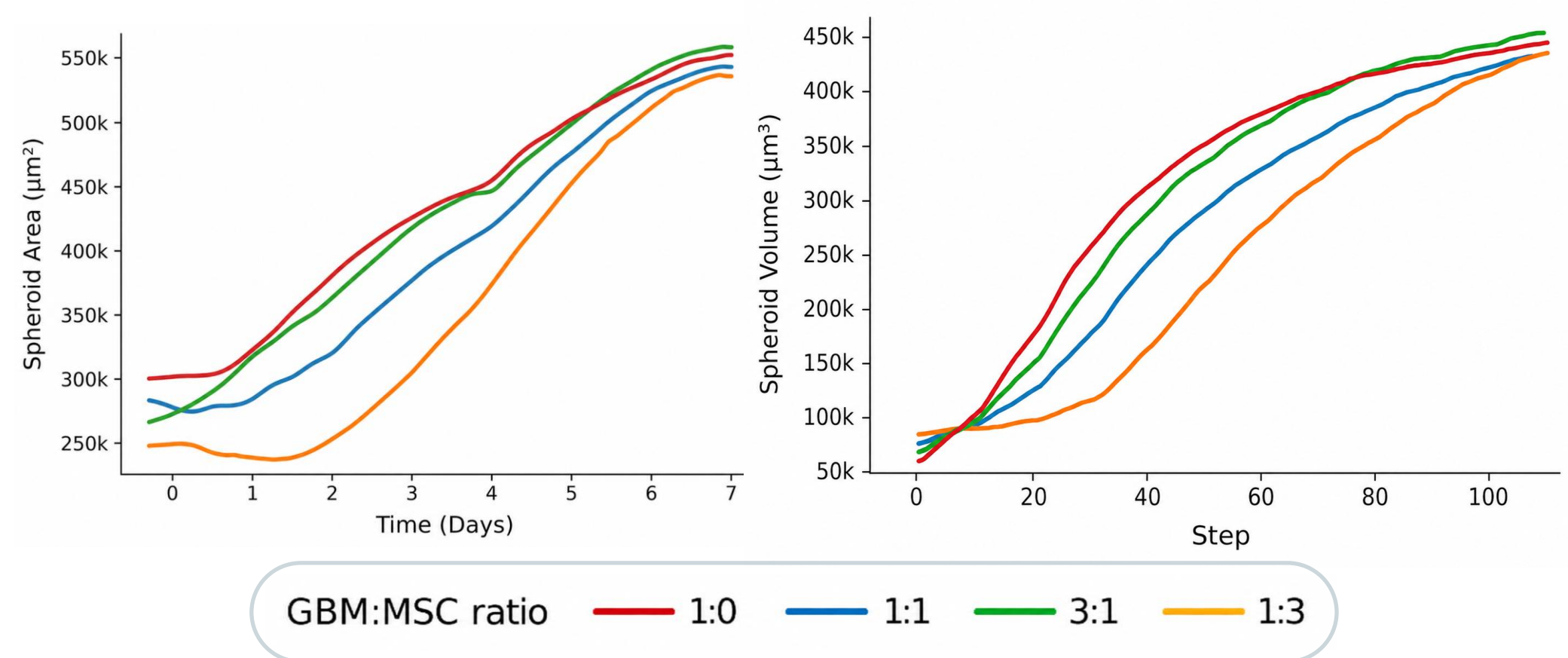
Emergent field gradients



State zonation

4 PRO-TUMOR MSC PHENOTYPE NEEDED TO REPRODUCE EXPERIMENTAL TRENDS

MODEL-EXPERIMENT AGREEMENT



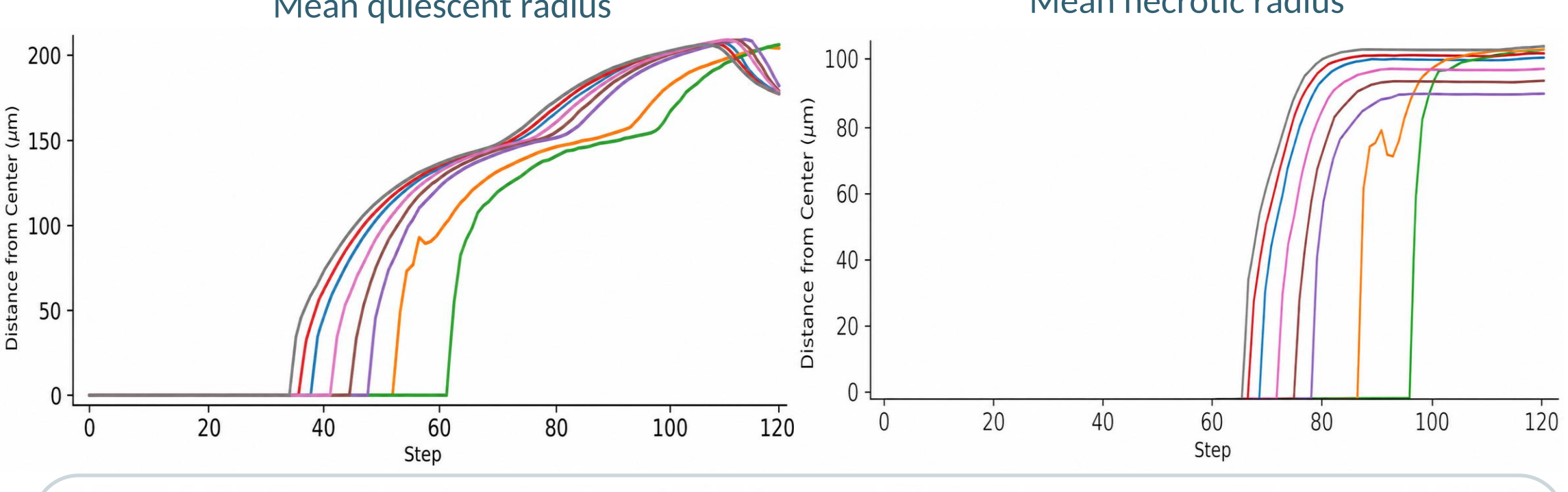
Qualitative agreement in curve ordering and overall trajectory
Comparison is qualitative (2D projected area vs 3D simulated volume)

PRO-TUMOR MSC RULES

- TRIGGERED BY**
 - high lactate / acidity
 - hypoxia
 - GBM contact
- MODELED ACTIONS**
 - growth-signal secretion
 - metabolic support
 - indirect buffering of stress / field gradients
- RESULT**
 - qualitative match to co-culture growth trends
 - delayed quiescent / necrotic core onset
 - sustained GBM proliferation

MODEL HYPOTHESIS

Core formation dynamics



Interpretation

- Higher MSC fractions delayed the first appearance of stressed core states.
- This suggests transient buffering of metabolic stress.
- Once buffering was exhausted quiescent and necrotic regions expanded rapidly.

5 SUMMARY

MSC EFFECT

GBM-MSC growth was **non-linear**: mixed spheroids could not be explained by pure GBM + pure MSC behavior alone.

MODEL

3D hybrid model captured **spheroid physiology**: viable rim, nutrient/metabolite gradients, and quiescent/necrotic core formation.

MECHANISTIC INTERPRETATION

Qualitative agreement required **MSCs** to act as context-dependent **pro-tumor** modifiers of the local microenvironment.

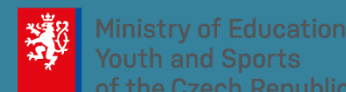
MODEL PREDICTION

MSC-rich spheroids may **delay** metabolic stress, then **accelerate** quiescent/necrotic core expansion.

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Any questions? Contact me!

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More info on my Github:

